



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

KARAN DAVE, 26070122507, CSE B3, 2025-2029

ASSIGNMENT No: 10

TITLE: Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.



Exercise

1. Create an abstract Payment class and implement Credit Card and UPI payment classes?

```
1 abstract class Payment {
2     double amount;
3     Payment(double amount) {
4         this.amount = amount;
5     }
6     abstract void makePayment();
7     void displayAmount() {
8         System.out.println("Payment Amount: ₹" + amount);
9     }
10 }
11 class CreditCard extends Payment {
12     String cardNumber;
13     CreditCard(double amount, String cardNumber) {
14         super(amount);
15         this.cardNumber = cardNumber;
16     }
17     void makePayment() {
18         System.out.println("Payment made using Credit Card.");
19         System.out.println("Card Number: " + cardNumber);
20     }
21 }
22 class UPI extends Payment {
23     String upiId;
24     UPI(double amount, String upiId) {
```

----- Credit Card Payment -----
Payment Amount: ₹5000.0
Payment made using Credit Card.
Card Number: 1234-5678-9012-3456

----- UPI Payment -----
Payment Amount: ₹250000.0
Payment made using UPI



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```
24     UPI(double amount, String upiId) {
25         super(amount);
26         this.upiId = upiId;
27     }
28     void makePayment() {
29         System.out.println("Payment made using UPI.");
30         System.out.println("UPI ID: " + upiId);
31     }
32 }
33 public class Main {
34     public static void main(String[] args) {
35         CreditCard creditCard = new CreditCard(5000, "1234-5678-9012-3456");
36         UPI upi = new UPI(250000, "karan@upi");
37         System.out.println("----- Credit Card Payment -----");
38         creditCard.displayAmount();
39         creditCard.makePayment();
40         System.out.println();
41         System.out.println("----- UPI Payment -----");
42         upi.displayAmount();
43         upi.makePayment();
44     }
45 }
```

Problems | Javadoc | Declaration | Console | Properties

<terminated> Main (1) [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (15-Aug-2026, 5:30:58 pm)

```
----- Credit Card Payment -----
Payment Amount: ₹5000.0
Payment made using Credit Card.
Card Number: 1234-5678-9012-3456

----- UPI Payment -----
Payment Amount: ₹250000.0
UPI ID: karan@upi
```

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```
----- Credit Card Payment -----
Payment Amount: ₹5000.0
Payment made using Credit Card.
Card Number: 1234-5678-9012-3456
```

```
----- UPI Payment -----
Payment Amount: ₹250000.0
Payment made using UPI.
UPI ID: karan@upi
```



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2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

```
FoodOrder.java
1  abstract class FoodOrder {
2      double foodPrice;
3      FoodOrder(double foodPrice) {
4          this.foodPrice = foodPrice;
5      }
6      abstract void calculateBill();
7  }
8  class DineInOrder extends FoodOrder {
9      double serviceCharge;
10     DineInOrder(double foodPrice, double serviceCharge) {
11         super(foodPrice);
12         this.serviceCharge = serviceCharge;
13     }
14     void calculateBill() {
15         double total = foodPrice + serviceCharge;
16         System.out.println("Order Type: Dine-In");
17         System.out.println("Food Price: ₹" + foodPrice);
18         System.out.println("Service Charge: ₹" + serviceCharge);
19         System.out.println("Total Bill: ₹" + total);
20     }
21 }
22 class TakeAwayOrder extends FoodOrder {
23     double packingCharge;
24     TakeAwayOrder(double foodPrice, double packingCharge) {
25         super(foodPrice);
26         this.packingCharge = packingCharge;
27     }
28     void calculateBill() {
29         double total = foodPrice + packingCharge;
30         System.out.println("Order Type: Take-Away");
31         System.out.println("Food Price: ₹" + foodPrice);
32         System.out.println("Packing Charge: ₹" + packingCharge);
33         System.out.println("Total Bill: ₹" + total);
34     }
35 }
```

Problems | Javadoc | Declaration | Console | Properties

<terminated> FoodOrder [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (15-Aug-2026, 5:45:07)

Service Charge: ₹50.0
Total Bill: ₹550.0

----- TAKE-AWAY ORDER -----
Order Type: Take-Away
Food Price: ₹500.0
Packing Charge: ₹30.0
Total Bill: ₹530.0

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```
24 TakeAwayOrder(double foodPrice, double packingCharge) {
25     super(foodPrice);
26     this.packingCharge = packingCharge;
27 }
28 void calculateBill() {
29     double total = foodPrice + packingCharge;
30     System.out.println("Order Type: Take-Away");
31     System.out.println("Food Price: ₹" + foodPrice);
32     System.out.println("Packing Charge: ₹" + packingCharge);
33     System.out.println("Total Bill: ₹" + total);
34 }
35 }
36 class Main2 {
37     public static void main(String[] args) {
38         DineInOrder dineIn = new DineInOrder(500, 50);
39         TakeAwayOrder takeAway = new TakeAwayOrder(500, 30);
40         System.out.println("----- DINE-IN ORDER -----");
41         dineIn.calculateBill();
42         System.out.println();
43         System.out.println("----- TAKE-AWAY ORDER -----");
44         takeAway.calculateBill();
45     }
46 }
```

Problems Javadoc Declaration Console Properties
<terminated> FoodOrder [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (15-Aug-2026, 5:45:07)
Service Charge: ₹50.0
Total Bill: ₹550.0

----- TAKE-AWAY ORDER -----
Order Type: Take-Away
Food Price: ₹500.0
Packing Charge: ₹30.0
Total Bill: ₹530.0

----- DINE-IN ORDER -----

Order Type: Dine-In
Food Price: ₹500.0
Service Charge: ₹50.0
Total Bill: ₹550.0

----- TAKE-AWAY ORDER -----

Order Type: Take-Away
Food Price: ₹500.0
Packing Charge: ₹30.0
Total Bill: ₹530.0